Comparative Evaluation of Low-Cost Sensing and Mapping Solutions for Teleoperated Ground Robots in USAR

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Abstract. Reliable and affordable ground robotic platforms for Urban Search and Rescue (USAR) remain a critical challenge, especially for cost-constrained deployments in public safety and defense. This paper presents an architecture for teleoperated ground robots designed for irregular terrain, developed through the systematic evaluation of industry-established sensing and mapping solutions. A comparative hardware study was conducted for 3D environment mapping, contrasting depth cameras (Intel RealSense D435 vs. Logitech C920e) and LiDAR sensors (Ouster vs. LDS-02), alongside five SLAM and depth estimation algorithms: Ouster Spherical Range Projection, Cartographer, ORB-SLAM3, MiDaS, and MapAnything. Dimensional reconstruction errors were measured against physical references: the Intel RealSense D435 achieved a 2.11% average error in a room scenario, MapAnything reached 5.50%, and MiDaS reached 15.17%. The LDS-02 combined with Cartographer matched the Ouster LiDAR's 2D mapping accuracy (6.24% vs. 5.74% dimensional error) at a fraction of the cost. ORB-SLAM3 was effective for sparse visual odometry but presented inherent monocular depth ambiguities. A hybrid track-and-leg locomotion system was developed in-house to traverse complex terrains. The resulting architecture demonstrates that cost-effective combinations of sensors and algorithms can meet USAR operational requirements. Current validation is bounded to laboratory and competition environments, with outdoor field trials established as the primary next step.

Keywords: USAR robotics · SLAM evaluation · teleoperation · sensor comparison · rescue robotics · low-cost robotics

1 Introduction

Urban Search and Rescue (USAR) operations impose extreme physical and operational risks on human responders. Ground robotic platforms capable of navigating debris-filled environments, generating real-time environment maps, and executing tasks offer a critical complement to traditional rescue methods [1]. However, the high cost of commercial robotic systems for USAR significantly

limits their large-scale deployment, especially in resource-constrained contexts such as public safety agencies and national defense. A fundamental open challenge is determining which sensing and mapping technologies provide adequate performance for USAR at an acceptable cost. A wide range of sensors and algorithms is available, from depth cameras and high-performance LiDAR systems to low-cost webcams and 2D laser scanners, yet systematic evaluations comparing their cost-performance trade-offs are scarce. Likewise, designs focused not only on performance but also on affordable production remain underexplored. This paper addresses these challenges through a comparative evaluation of industry-established sensing, mapping, and manipulation technologies, integrated into a teleoperated ground robot architecture. The work was carried out by the RoboIME team of the Military Institute of Engineering (IME), Brazil, in the context of the RoboCup Rescue Robot League 2026, which served as the benchmark for the evaluation. The team has prior institutional experience in military robotics and autonomous mapping. The main contributions of this paper are:

- A quantitative cost-performance evaluation comparing two camera modalities (Intel RealSense D435 vs. Logitech C920e) and two LiDAR sensors (Ouster vs. LDS-02) for environment mapping in USAR contexts;
- A systematic empirical characterization of five mapping and depth algorithms (Ouster Spherical Range Projection, Cartographer, ORB-SLAM3, MiDaS, and MapAnything), covering reconstruction accuracy, computational footprint, memory usage, and pipeline latency on low-cost platforms;
- A design study of a modular hybrid track-and-leg locomotion platform, establishing the architecture and manufacturing constraints for future low-cost USAR field deployments.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the system architecture. Section 4 presents the evaluation methodology. Section 5 reports and discusses the results. Section 6 concludes the paper.

2 Related Work

Research on the use of robotics for USAR gained momentum in the 21st century. Murphy [1], in her seminal work on disaster robotics, systematizes the history of the use of these robots in real operations and the lessons learned. Ground robots for USAR typically require three core capabilities: terrain-adaptive locomotion, environment mapping, and manipulation for interaction with victims. For environment mapping in mobile robots, SLAM is the fundamental technique. Beyond individual techniques, standardization efforts have been fundamental for objectively evaluating USAR robots. The *RoboCup Rescue Robot League* (RRL) emerged from an initiative of the *National Institute of Standards and Technology* (NIST) to create reference test environments and, since its incorporation into RoboCup in the early 2000s, has become one of the leading international

testbeds for rescue platforms. The competition employs standardized arenas that reproduce partially collapsed buildings, with obstacles based on the *DHS-NIST-ASTM International Standard Test Methods for Response Robots*, such as the *stepfields*, used since 2005 to evaluate mobility on irregular terrain. By gathering, within a single controlled environment, the three core requirements of USAR robots (adaptive locomotion, mapping, and manipulation), the RRL offers a representative evaluation scenario, and is adopted in this work as the testbed for comparing the studied sensing, mapping, and manipulation solutions.

3 System Architecture

3.1 Locomotion Platforms

Three ground platforms were studied, representing a progression in cost and capability, as part of the team's strategy to identify the best balance between cost and performance.
TurtleBot3 [2]: the initial study platform. Despite its limited capability, it provided fundamental experience with ROS 2 and LiDAR-based mapping, serving as a testbed for the first experiments (Fig. 1).

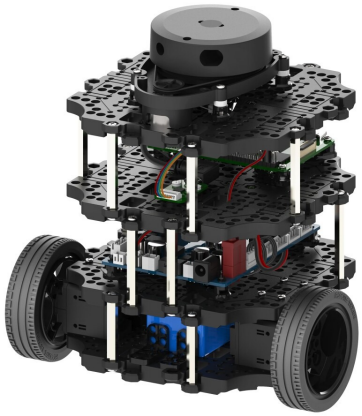


Fig. 1. TurtleBot3

Tri-Track Rover: the main locomotion study platform for irregular terrain. Its track system offers high terrain adaptability; this model, together with the study of the platforms used in the RRL, was used for the development of a modular hybrid platform that allows the central body to be separated from the locomotion modules, switching between the track and leg systems. The platform is in the phase of analyzing manufacturing methods, focusing on additive manufacturing as a low-cost production method.

Clearpath Jackal (Fig. 2): a higher-cost reference platform, used in full system integration tests and in higher-fidelity mapping experiments, due to its compatibility with and support for the Ouster LiDAR. Its onboard IMU, 270 Wh lithium battery, and 4WD drive system provide a stable base for sensor benchmarking (Table 1).

Table 1. Hardware Components — Clearpath Jackal

Part	Brand & Model
Drive motors (4)	Integrated 4WD
Motor drivers	Integrated 32-bit MCU
DC/DC	User power supply (5 V, 12 V, 25.6 V)
Battery management	Integrated protection
Battery	AllCell Technologies Li, 270 Wh
Microcontroller	32-bit MCU
Computing unit	Intel- or Jetson-based
Wi-Fi adapter	Integrated
IMU	Onboard IMU
Battery charger	110–220 VAC charger

3.2 Sensor Suite

Two camera modalities were evaluated for 3D mapping:

- *Intel RealSense D435*: an RGB-D camera widely used in USAR robotics, representing the state of the art for embedded 3D mapping in its price range.
- *Logitech C920e*: a standard monocular webcam evaluated as a low-cost alternative, using software-based depth estimation instead of hardware depth sensing.

Two LiDAR sensors were evaluated:

- *Ouster*: a high-performance LiDAR that enables the precise mapping of large areas when combined with the Jackal platform.
- *360 LDS-02 Laser Distance Sensor*: a low-cost 2D LiDAR that provides satisfactory local mapping and operator guidance, used on the TurtleBot3.

3.3 Mapping Algorithms Tested

Five algorithms were evaluated under the available sensor configurations. Two of them operate on the LiDAR, generating 2D occupancy grid maps and:



Fig. 2. Clearpath Jackal

1. *Ouster Spherical Range Projection*: deterministic conversion from spherical to Cartesian coordinates, per-beam calibrated, native to the Jackal. Each laser shot returns a distance (*range*); combined with the encoder's rotation angle and the intrinsic azimuth and elevation angles of each channel (obtained via the sensor's factory calibration), the driver projects each measurement into a point (x, y, z) in the sensor's frame of reference. It is not a SLAM algorithm, it is the pre-processing stage that feeds both the *scan matching* and the obstacle detection.
2. *Cartographer* [4]: pose-graph-based SLAM. Scans are accumulated into local submaps (*probability grids*) through scan matching solved with non-linear optimization (Ceres); upon completing each submap, the system searches for loop closures between submaps and globally optimizes the pose graph, correcting accumulated drift. It offers superior consistency to raw mapping, at the cost of higher computational demand, while still being considered lightweight, being able to run on a *Raspberry Pi 4*, for example.

The remaining three use a monocular camera for 3D reconstruction, from the sparse feature map to the dense point cloud:

1. *ORB-SLAM3* [5]: feature-based visual (and visual-inertial) SLAM, which extracts ORB descriptors for tracking, mapping, and relocalization. It builds a *sparse* map of 3D points organized by a covisibility graph and refined by *bundle adjustment*, and introduces the *Atlas* multimap system, which starts a

Table 2. Consolidated Comparison of Average Sensor Costs (US vs. Brazil Market)

Sensor	Modality / Technology	Average Price (USD)	Average Price (BRL)	Sources / Research Locations
Logitech C920e	Monocular Webcam (Software-based Depth)	US$75.00	R$396.50	KaBuM!, STI Computadores, Shopee (BR), Walmart (US)
ROBOTIS LDS-02	2D LiDAR, 360°	US$189.00	R$2,150.00	ROBOTIS Official, RobotShop, Robotics Retailers (BR)
Intel RealSense D435	RGB-D Camera	US$324.00	R$5,095.00	TendInfo, Mercado Livre (BR), Intel Store, Mouser
Ouster LiDAR (OS1)	Industrial 3D LiDAR	US$7,250.00	R$67,500.00	Ouster Inc., Geotechnical Distributors (BR)

new submap when tracking is lost and merges it upon recognizing previously visited areas. Evaluated with the Logitech C920e.

2. *MiDaS* [6]: a deep-learning network for monocular depth estimation, trained with a mixture of multiple datasets to enable *zero-shot* transfer across domains. From a single RGB image, it predicts a dense depth map, reprojectable into a dense point cloud. Since each frame is processed independently, the method does not enforce consistency between frames.
3. *MapAnything* [7]: a *feed-forward transformer* model for metric 3D reconstruction, which takes one or more images (optionally with intrinsic parameters or poses) and directly regresses the metric 3D geometry and the cameras, through a factored representation (depth maps, local *ray maps*, poses, and a metric scale factor). It produces dense, multi-frame-consistent reconstruction in a single forward pass, being the most recent method evaluated and the one with the highest computational demand.

Cartographer was natively integrated into the ROS 2 stack of the TurtleBot3 platform, using the LDS-02 sensor as the data source. *ORB-SLAM3* was obtained directly from the official repository, requiring an update of the build system to C++17, through CMake configuration, to ensure compatibility with the development environment used. The algorithm was run standalone, without integration into the ROS 2 stack, using the Logitech C920e camera as input. The *MiDaS* and *MapAnything* algorithms were integrated into the ROS 2 stack

through a node architecture developed by the team. For each algorithm, two nodes were created: a camera node, responsible for capturing and publishing the Logitech C920e images, and an inference node, responsible for receiving the images, processing the depth map through the deep-learning model, and publishing the result on a ROS 2 depth topic. The final mapping was carried out by RTAB-Map, which consumed the image and depth topics published by the inference nodes to generate the 3D point cloud.

Table 3. Software

Name	Version	License	Use
Ubuntu	20.04	Open	OS
ROS 2	Jazzy	Apache 2.0	Middleware
Cartographer	—	Apache 2.0	2D/3D SLAM
ORB-SLAM3	3	GPLv3	Visual SLAM
MiDaS	—	MIT	Monocular depth
MapAnything	—	—	3D reconstruction

4 Evaluation Methodology

The evaluation strategy follows a systematic cost-performance comparison approach. Each sensing module was first compared against the native systems of the established models. Then, they were evaluated independently, in their intended configuration, listing their main characteristics, advantages, and shortcomings, seeking possibilities of use in different applications. Consequently, the evaluation prioritized algorithmic characterization, identifying capabilities replicable on any low-cost monocular camera and LiDAR while maintaining adequate quality for field operations.

4.1 Evaluation of Locomotion Platforms

The evaluation of the locomotion platform was carried out by reusing other projects focused on ground platforms and projects already established in the RRL. Since these platforms already had extensive studies, the project sought only to define the type of locomotion it intended to develop by comparing the results of those studies with the needs and functions intended for the final product, without actually carrying out a specific test.

4.2 Evaluation of SLAM and Depth Estimation

The evaluation of the algorithms was conducted in two phases. In the first, a direct cost-performance comparison was carried out between the algorithms tested

and the established reference maps, generated by the Jackal platform with high-fidelity sensors. At this stage, the following criteria were evaluated: coverage of the mapped area, geometric consistency, scan stability, and loop closure on repeated trajectories.
To measure geometric consistency, 3 test environments were set up to evaluate the different methods. The first consisted of a corridor path used to evaluate the Ouster and LDS-02 LiDARs with the Cartographer algorithm using 2D maps. The second environment is based on a table with items distinct in shape, color, and size, ranging between 2.5 cm and 45 cm, focusing on the camera algorithms to measure the level of precision and the influence of the items' shape and colors on the measurement. Finally, the laboratory room itself was mapped to evaluate the capacity for reconstruction of enclosed environments, and key objects had their size measured on the map in order to compare the precision for objects at a larger scale, both in the size of the objects and in the distance to the camera.

The root mean square error (RMSE) measures the typical magnitude of the deviation between the estimated values m_i and the reference values r_i over n samples, being defined as:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(m_i - r_i)^2} \tag{1}$$

By squaring each residual before averaging, the metric prevents cancellation between deviations of opposite signs and weights errors of greater magnitude more heavily, while the final square root restores the original unit of the measured quantity. This sensitivity is desirable when large errors are operationally more critical than small ones, but it makes the indicator vulnerable to *outliers*, which is why it is reported here together with the median of the residuals.

In addition to the qualitative evaluation of the depth maps, the computational performance of the algorithms was analyzed through systematic monitoring of resources and execution times during 3 minutes of execution of each algorithm. The metrics tracked included the image publishing rate (FPS), the node latency, the total pipeline latency, and the consumption of CPU, GPU, RAM, and VRAM. Additionally, the temporal distribution of the acquisition, pre-processing, inference, post-processing, and publishing stages was mapped, evidencing the impact of each stage on the total system time. In the second phase, each algorithm was characterized individually, without the goal of a direct comparison but highlighting its operational advantages and limitations, robustness under adverse conditions (low visual information, irregular terrain), and computational demand, determining factors for the feasibility of deployment on low-cost platforms.

5 Results and Discussion

5.1 Locomotion Platforms

The initial study of existing locomotion platforms shows 3 main systems, wheels, tracks, and legs, related to the criteria of speed and obstacle-traversal capability. The analysis of the mechanisms studied showed that these capabilities are relatively mutually exclusive, and it was concluded that the best path would be to seek a balance between the two. Therefore, after analyzing the RRL projects, the development of a hybrid leg-and-track platform was initiated. The platform is based on 4 track modules that are independent of each other and capable of moving their tracks but also of moving the modules as a whole, allowing them to act as legs.

5.2 LiDAR Algorithms

The results obtained with the two LiDAR-based algorithms revealed complementary performance profiles, with distinct advantages and limitations depending on the application scenario. *Cartographer* produced satisfactory 2D occupancy grid maps in small-scale indoor environments. Compared to the reference map, the coverage of the mapped area was adequate, but geometric consistency degraded over repeated trajectories: upon revisiting the same path, scan overlaps appeared that generated structure duplications in the map; real-time loop closure attempted to correct the accumulated drift over extensive trajectories, producing coherent maps but only when moved at low speed (around 0.1 m/s). This behavior reflects the limitation of 2D SLAM when encountering rapid dynamic changes without a robust global loop-closure mechanism.
In contrast, the mapping tests generated by the *Jackal* presented, in addition to adequate instantaneous mapping, a consistent map-to-map association even at higher linear speeds (1 m/s). However, residual association errors could be seen after movements with high angular velocity (2.0 rad/s), which shows a not-fully-solved problem, as with the native mapping algorithm.
In operational terms, the two sensors have a significant difference in that the LDS-02 LiDAR is a 2D mapping sensor and the Ouster is a 3D sensor, a factor that likely facilitated the submap association due to the greater data coverage. In contrast, analyzing the percentage errors of the map measurements, it is possible to identify that, for the 2D environment, both have very similar precision. Thus, the experiment validates that, specifically for the criteria of 2D mapping, the solution of using the LDS-02 combined with Cartographer produces a satisfactory solution that is considerably cheaper than the Ouster LiDAR. The negative factors perceived are the limitation to 2D maps for the LDS-02 LiDAR, but this aspect can be overcome by combining it with a monocular camera for mapping, or ignored if it is not necessary for the project's scope, and, mainly, the speed limitation; the use of other methods to support map association was studied, such as a better integration with the IMU or the limitation of distant measurements to use only higher-precision data, but it was considered that the

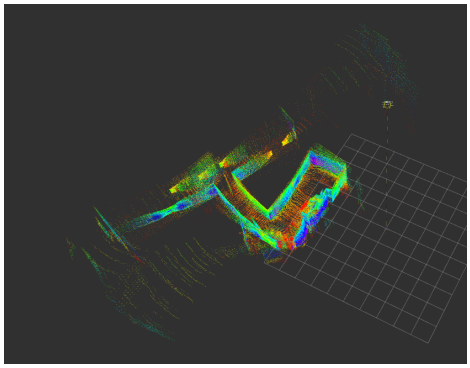


Fig. 3. 3D point cloud map generated by the Jackal platform using the Ouster LiDAR.

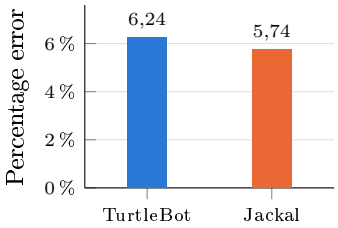


Fig. 4. Percentage error by robotic platform.

application of these mechanisms would fall outside the scope of this analysis, since one would no longer be dealing with the algorithm purely.

These error percentages represent initial point estimates; a formal measurement uncertainty budget following the GUM framework (ISO/IEC Guide 98-3) is designated for future metrological characterization.

5.3 Monocular Camera Algorithms

The results obtained with the three monocular-camera-based algorithms revealed distinct performance profiles, with complementary advantages and limitations depending on the reconstruction requirement considered. *ORB-SLAM3* (Fig. 5) demonstrated robustness for visual odometry and sparse mapping, maintaining tracking under conditions of varied texture. Compared to the reference map, the spatial coverage was satisfactory for navigation, but the accuracy of the depth estimation proved extremely limited: the feature points were systematically positioned closer than their actual positions, and the geometry of the sparse map did not consistently reflect the actual shape of the environment, which prevented measurement in the high- and low-scale tests. Individually, the algorithm proved suitable for localization and visual odometry in environments with sufficient texture; on surfaces with low visual variation, such as white walls, feature tracking was impaired, with recurrent tracking losses that led the algorithm to restart the

mapping, compromising the continuity and coherence of the global map. *MiDaS*

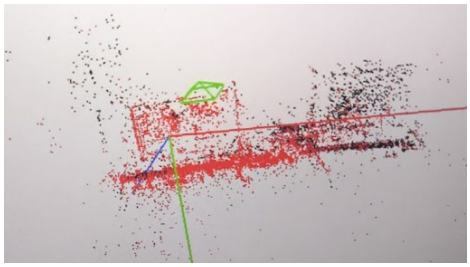


Fig. 5. Sparse map generated by ORB-SLAM3 using the Logitech C920e monocular camera. Note the depth inconsistencies visible in the distribution of the points.

(Fig. 6) produced dense point clouds from monocular input, filling in the local geometry per frame with greater detail than feature-based methods. Compared to the reference map, the per-frame reconstruction density was satisfactory, evidencing the method's potential for detailed capture of local geometry at high scale. In contrast, in the low-scale measurement (items placed on the table), the method was unable to build a consistent point cloud, which prevented its measurement. Individually, the absence of correspondence between consecutive frames introduced temporal inconsistency, generating disconnected planes along the sequence, visible in Fig. 6, and compromising the coherence of the global 3D reconstruction. This behavior makes the method more suitable for static point inspection of scenes than for continuous mapping during navigation.

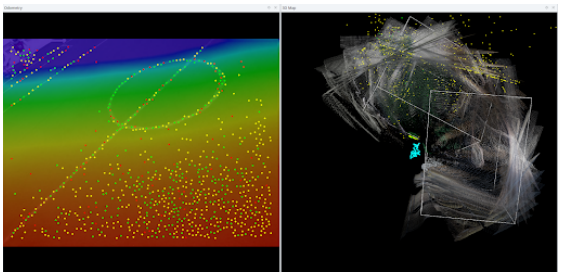


Fig. 6. Dense point cloud generated by MiDaS from a sequence of monocular images. The disconnected planes (visible as gaps between surface segments) result from the lack of correspondence between frames.

MapAnything (Fig. 9) produced dense 3D reconstruction that was geometrically consistent across multiple frames, establishing geometric correspondences in a single forward pass that preserve spatial coherence along the sequence,

without the formation of disconnected planes. Compared to the reference map, the global geometric consistency was satisfactory, approaching the metric precision of the industry-validated model with the Intel RealSense at both scales, which makes the model suitable for field mapping applications. Individually, the

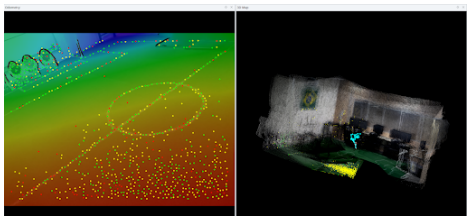


Fig. 7. 3D reconstruction generated by MapAnything in the native visualization.

method presents as a limitation the geometric fidelity in the point-cloud export: when visualizing the map in Meshlab (a tool for visualizing and processing 3D meshes) (Fig. 10), the resulting geometry presented localized structural distortions, such as curvature of flat surfaces and loss of proportionality of the environment, not observed in the native visualization of the reconstruction, where the metric proved satisfactory compared to the real environment. The main negative factor is a higher computational cost compared to the other algorithms; however, the values were not considered catastrophic to the point of requiring a very high-cost processor.

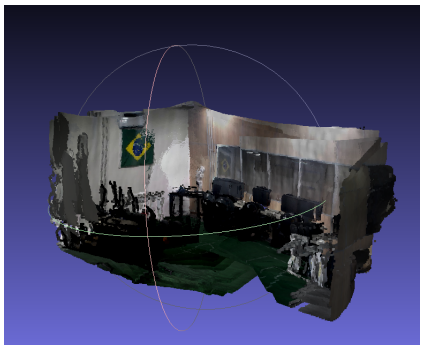


Fig. 8. . Point cloud generated by MapAnything visualized in Meshlab.

5.4 Comparative Synthesis

The previous sections presented the visual behavior of each computer vision algorithm based on the processed images. To guide the choice of the most suitable

method for a USAR robot, however, it is necessary to cross-reference these measurements within a single frame of reference. Table 4 summarizes this qualitative synthesis, whose criteria (sensor cost, computational cost, map density, and geometric consistency) are justified quantitatively by the charts presented below. The qualitative classifications presented in Table 4 (in particular computational

Table 4. Comparison of the evaluated SLAM and depth estimation algorithms

Algorithm	**Sensor**	**Sensor cost**	**Comp. cost**	**Map density**	**Geometric consistency**
Cartographer	2D LiDAR	Medium	Medium	2D	High
ORB-SLAM3	Monocular camera	Low	Low	Sparse 3D	Low
MiDaS	Monocular camera	Low	Medium	Dense 3D	Low-Medium
MapAnything	Monocular camera	Low	High	Dense 3D	Medium-High

cost, map density, and geometric consistency) are not arbitrary: each one derives directly from the throughput (FPS), memory usage, *pipeline* latency, and reconstruction error measurements detailed below. Fig. 9 establishes the first criterion, that of computational cost, comparing the capture rate of each sensor (input FPS) with the rate effectively processed (output FPS): the greater the drop between these two values, the higher the computational cost of the method relative to the available hardware. This comparison already separates

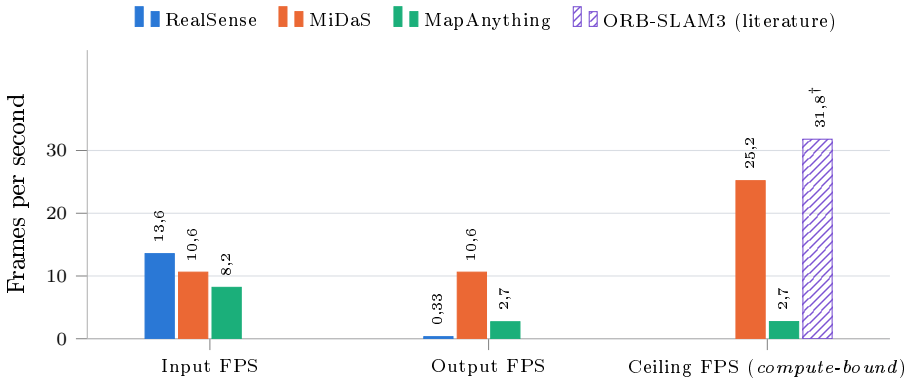


Fig. 9. FPS performance analysis of the evaluated systems. The chart contrasts the capture rates (input), effective processing (output), and the maximum theoretical processing limit (ceiling) for the different approaches

the three cost profiles of Table 4. *MiDaS* keeps the output rate identical to the input rate (10.6 FPS), processing each frame in real time without accumulating delay, which justifies its *Medium* cost classification. *MapAnything*, in turn, suffers a drop of approximately 3× (from 8.2 to 2.7 FPS), already operating at its processing ceiling (2.73 FPS *compute-bound*), which confirms its *High* cost. *ORB-SLAM3*, evaluated from literature data, sustains a ceiling of 31.8 FPS by relying on sparse *feature* tracking instead of dense per-frame reconstruction, hence its *Low* cost. Throughput, however, says nothing about the geometric quality of the map generated. Fig. 10 completes this criterion by measuring the percentage reconstruction error in two scenarios of distinct scale. The two sce-

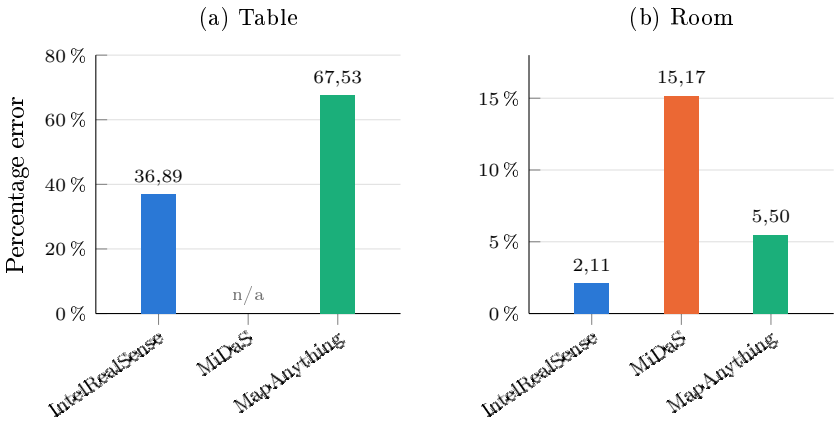


Fig. 10. Percentage error by method in the evaluated scenarios. The vertical scales are independent between the panels; *n/a* indicates absence of measurement.

narios reveal a consistent pattern: *MapAnything* errs more than the RealSense depth camera in both cases (67.53% against 36.89% in the Table scenario; 5.50% against 2.11% in the Room scenario), and *MiDaS* presents the highest relative error in the Room scenario (15.17%), with no measurement available in the Table scenario, since the point cloud did not present sufficient detail for measuring the reference objects. This result supports the geometric consistency classification of Table 4: *MapAnything* at *Medium-High* (superior to MiDaS, but still far from the direct-depth reference) and *MiDaS* at *Low-Medium*, penalized by the absence of a temporal constraint between consecutive frames. Computational cost and precision, however, do not exhaust the practical feasibility of each method for an embedded USAR robot: it remains to evaluate memory consumption, shown below.

The pattern of average *versus* peak RAM reinforces the reading of computational cost already obtained via FPS. *MiDaS* keeps RAM practically constant between average and peak (1353 MB and 1354 MB), a sign of a stable memory footprint, compatible with continuous-stream processing. *MapAnything* also

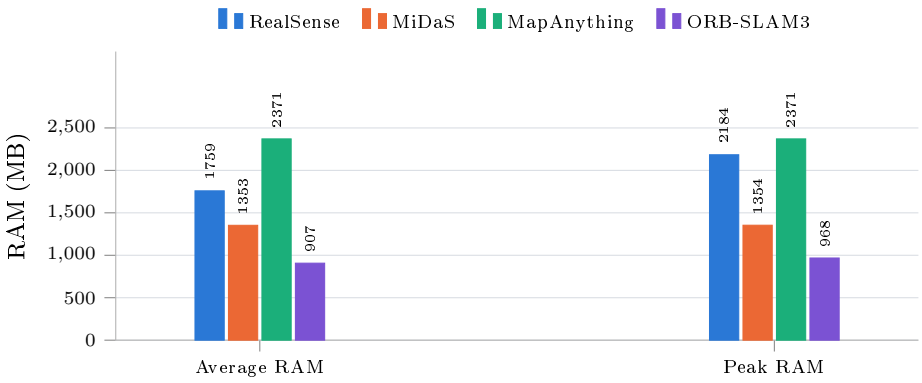


Fig. 11. Average and peak RAM consumption of the evaluated methods during three minutes of continuous execution.

shows an average equal to the peak (2371 MB), but at a much higher level; there is no headroom between typical operation and maximum-load operation, indicating that the method already operates at the upper limit of its memory consumption in a sustained manner. *ORB-SLAM3* confirms its computational lightness with the lowest consumption among all methods (907 MB average, 968 MB peak), consistent with maintaining a sparse map instead of a dense point cloud. The joint analysis evidences that there is no universally superior solution: each algorithm presents a distinct profile that makes it more suitable depending on the operational requirements of the mission and the constraints of the available hardware.

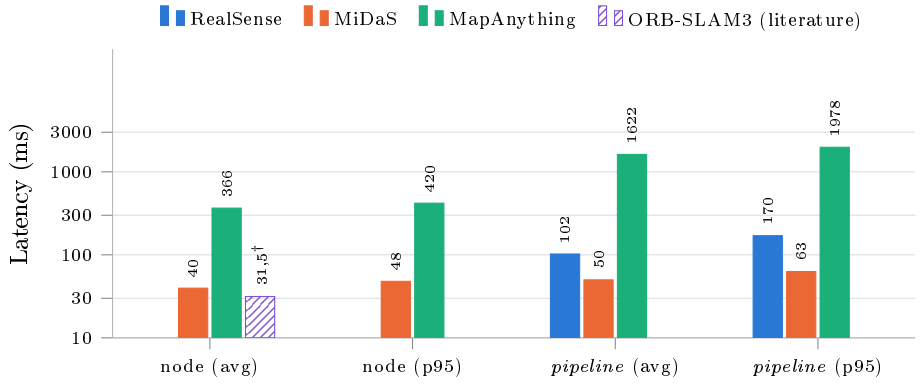


Fig. 12. Latency of the evaluated methods, decomposed into individual node time and full *pipeline* time (average and 95th percentile). The vertical axis is logarithmic. The hatched bar (†) reports a value taken from the literature, not measured in this setup; the *pipeline* latency of ORB-SLAM3 was not measured.

The latency analysis, decomposed into individual node time and full *pipeline* time, exposes exactly where the computational cost of each method is concentrated. In *MiDaS*, the difference between node (40 ms) and *pipeline* (50 ms) is small, which indicates that the integration cost is low and most of the time is spent on the inference itself. In *MapAnything*, on the other hand, the jump between node (366 ms) and *pipeline* (1622 ms) (an amplification of almost 4.5×) shows that the global map consolidation stage (and not the isolated per-frame inference) is the main bottleneck of the method, which reinforces its *High* cost classification and explains why its effective output rate in Fig. 9 falls so far below the input rate. *ORB-SLAM3*, with only 31.5 ms of node latency reported in the literature, once again evidences its lightweight profile, suitable for real-time tracking. This trade-off relationship between latency and memory becomes even clearer when positioning the four methods in the same two-dimensional space, as in the figure below: *ORB-SLAM3* alone occupies the region of lowest RAM consumption, since its full *pipeline* latency was not measured in this work; *MiDaS* concentrates in the quadrant of low latency and moderate RAM; and *MapAnything* occupies the opposite extreme, with a *pipeline* latency more than 15× higher than that of MiDaS and the highest memory consumption among the methods evaluated.

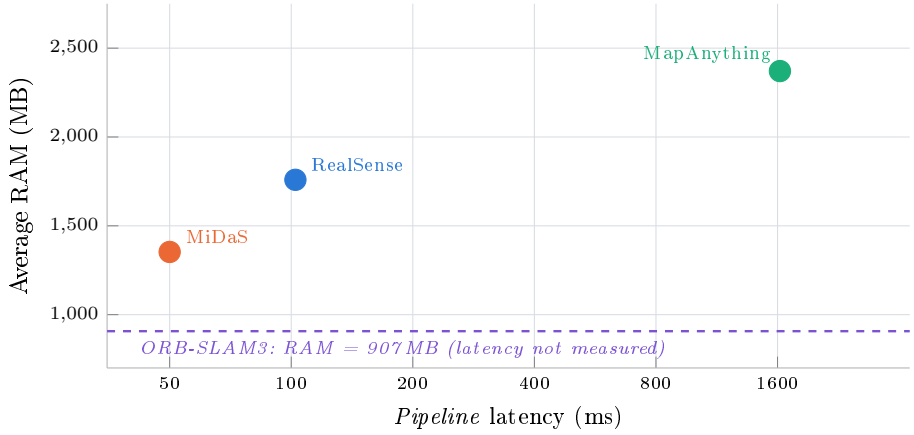


Fig. 13. Trade-off between *pipeline* latency and average RAM consumption for the evaluated methods. The horizontal axis is logarithmic. ORB-SLAM3 is shown as a dashed reference line because only its memory consumption was measured in this work.

Combining throughput, precision, memory, and latency, it is now possible to indicate the most suitable use scenario for each algorithm. When sensor cost is the dominant constraint and some 3D reconstruction is necessary, the monocular camera algorithms become viable alternatives. *ORB-SLAM3* is indicated for scenarios in which localization and visual odometry are prioritized over the metric fidelity of the map, such as tracking the robot's trajectory in environ-

ments with sufficient texture. *MiDaS* is more suitable for static point inspection of specific areas (such as the detailed analysis of a debris region before the robot's approach), leveraging its high per-frame reconstruction density without the need for temporal consistency. *MapAnything*, finally, is the most indicated when dense and globally consistent 3D reconstruction is the central requirement, accepting the higher computational demand as a trade-off, being particularly relevant for the detailed mapping of complex environments intended for rescue-route planning. Algorithm selection should be guided by the hierarchy of mission requirements (navigation versus reconstruction, real-time versus post-processing, embedded hardware versus remote processing) and by the budget availability for the sensor suite.

6 Conclusion

The results of the experiments presented here indicate that the approach of low-cost sensors combined with mapping algorithms, even with an increase in computational cost, is feasible. A common monocular webcam, combined with ORB-SLAM3, is capable of generating operationally useful maps without relying on dedicated depth hardware; although it did not yield positive results on the mapping side, it presented itself as an excellent candidate for spatial positioning through real-time image comparison with high processing speed. MiDaS produced denser point clouds, but with temporal inconsistencies that limit its direct application, making it unviable for the generation of large-scale maps but already enabling the execution of specific tasks that require 3D visualization. Although it was discarded as a large-scale mapping method, it proved to be a considerable advance in 3D reconstruction. Among the methods evaluated, MapAnything offered the most coherent multi-frame 3D reconstruction, while Cartographer proved the most reliable option for 2D-LiDAR-based operation. The modular architecture adopted, in which the locomotion base separates from the central body, allows progressive functional expansion without requiring complete replacement of the platform, which is central to the low-cost development proposal. The experiment focused on the direct comparison between the algorithms, avoiding the application of filters and other non-native methods that could have improved their performance; thus, the application of these tools could further improve these solutions, possibly even surpassing the high-cost sensors. All results reported here are bounded to controlled laboratory and competition environments. As natural developments of this study, we highlight the incorporation of quantitative evaluation metrics and validation under unstructured outdoor debris conditions. The work presented here establishes the foundation upon which these next stages will be built. Furthermore, a formal measurement uncertainty budget following the GUM framework is reserved for future metrological characterization.

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